NYC 311 Service Analytics

Project Documentation

Calendar Year 2024 | SQL + Power BI

This document describes the purpose, analytical workflow, dashboard structure, and interpretation guidelines for the NYC 311 Service Analytics project. It is intended to help reviewers understand how the analysis was organized and how the dashboard should be interpreted.

1. Project Overview

The project analyzes New York City 311 service requests created during calendar year 2024. It uses SQL for data preparation and analytical querying and Power BI to present service-demand patterns, agency workload, resolution-time measures, and data-quality indicators.

The Power BI report is built from the table/view named vw_nyc_311_2024_powerbi. The report contains four pages: Executive Overview, Service Demand Analysis, Operational Efficiency, and Data Quality & Insights.

2. Business Objectives

- Measure the overall volume and status distribution of 311 service requests.
- Identify patterns in request volume by month, borough, and complaint category.
- Compare agency workload and descriptive resolution-time patterns.
- Examine seasonal demand for Heat/Hot Water complaints.
- Assess completeness and consistency of status, closure-date, and resolution-time fields.
- Translate descriptive findings into practical reporting and operational-planning considerations.

3. Dataset Description

The analysis covers NYC 311 service requests for 2024. The project dataset contains approximately 3.46 million request records (the report's Total Requests measure has been configured to display the exact count).

Field / concept	Analytical use
created_date / created month	Request timing and monthly demand trends
borough	Geographic comparison across NYC boroughs
complaint_type	Service-request category analysis
agency_name	Agency workload and resolution-time comparisons
status	Request-status distribution and data-quality checks
closed_date	Closure-date completeness and status/date consistency
resolution_time_days	Descriptive resolution-time analysis and banding
season	Winter versus summer comparison for seasonal analysis

Note: Confirm the exact source dataset title, source URL, extraction date, and field definitions against the original downloaded dataset before publishing this document. Those source-specific details are intentionally not guessed here.

4. Data Preparation and Quality Review

The Power BI model uses the project table/view vw_nyc_311_2024_powerbi. The report includes checks for missing closure dates, status/closure-date mismatches, and missing resolution-time values.

- Requests marked Closed but missing a closure date are counted separately.
- Requests with a non-Closed status but a populated closure date are counted separately.
- Records with a nonblank resolution_time_days value are counted as valid resolution records.
- Resolution times are grouped into Under 1 day, 1–3 days, 3–7 days, 7–30 days, 30–90 days, 90+ days, and Missing.

These checks identify records that may require validation. A missing or inconsistent field does not, by itself, prove that the underlying service request was handled incorrectly; it indicates a limitation in the recorded data.

5. SQL and Analytical Workflow

1. Load and organize the 2024 service-request data in the SQL environment.
2. Review relevant fields used for time, geography, complaint category, agency, status, and closure.
3. Prepare the analysis-ready table/view used by Power BI: vw_nyc_311_2024_powerbi.
4. Use SQL queries to support data validation and analytical exploration.
5. Connect the prepared data to Power BI and define report measures for request counts, closure rates, resolution-time statistics, and data-quality counts.

Project note: Add the final SQL scripts, table definitions, join logic, and any specific cleaning/transformation rules to the SQL folder. This section describes the high-level workflow and should be updated to match the final scripts exactly.

6. Power BI Report Structure

Page 1 — Executive Overview

High-level KPIs, monthly request volume, borough distribution, top complaint categories, and request-status distribution.

Page 2 — Service Demand Analysis

Monthly demand by borough, borough and complaint-type slicers, Heat/Hot Water monthly demand, and winter-versus-summer comparisons.

Page 3 — Operational Efficiency

Agency workload, average resolution time by agency and borough, and selected HPD complaint-type comparisons.

Page 4 — Data Quality & Insights

Closure-date/status quality indicators, valid and missing resolution-record counts, resolution-time bands, and business findings.

7. Measures and Definitions

Measure	Definition

Total Requests	Count of service-request records included in the current filter context.
Closure Rate	A project-defined closure measure. Confirm its exact numerator and denominator in the DAX measure before publishing the metric definition.
Average Resolution Days	Average of resolution_time_days in the current filter context.
Median Resolution Days	Median of resolution_time_days in the current filter context.
Non-Closed Status Requests	Count of records whose status is not Closed, according to the measure's filter logic.
Missing Closure Date Requests	Count of records with a blank closed_date.
Closed Status Missing Date	Count of records with status = Closed and a blank closed_date.
Non-Closed With Closure Date	Count of records with a non-Closed status and a populated closed_date.
Valid Resolution Records	Count of records with a nonblank resolution_time_days value.

8. Interpretation and Limitations

- Request volume represents reported service demand; it is not a direct measure of the underlying incidence of a problem.
- Agency workload comparisons reflect assigned request volume and should not be interpreted as a ranking of agency performance.
- Average resolution time can be affected by outliers, complaint mix, and differences in case complexity.
- Borough comparisons may reflect differences in population, reporting behavior, and service-request mix.
- Missing or inconsistent status and date fields can affect the completeness of resolution-time reporting.
- The dashboard is primarily descriptive. It does not establish causal relationships or prove that one operational factor caused another.

9. Recommendations for Use

- Use the Executive Overview to establish the scale and broad distribution of service demand.
- Use the Service Demand page to explore geographic, category, and seasonal patterns.
- Use the Operational Efficiency page to identify workload and resolution-time patterns that may warrant further investigation.
- Use the Data Quality page to identify fields and records that should be validated before relying on resolution metrics.
- Before making resource-allocation decisions, supplement request counts and average resolution times with prior-period comparisons, case complexity, staffing, service-level targets, and outcome measures.

10. Project Files

Recommended project folder contents:

01_Business_Requirements — BRD and KPI Definition Register.

02_Dataset — source and prepared datasets.

03_SQL_Analysis — SQL scripts and analysis.

04_PowerBI — PBIX and PDF.

05_Documentation — Project Documentation and supporting notes.

06_Final_Deliverables — Executive Summary, findings and recommendations.

07_Screenshots — four dashboard screenshots.